



Presented by **ทีมนักขุดทอง**

Gold Signal Intelligence via LLM Decision & ReAct Reasoning

Data Science for Signal Processing

Meet Our Team



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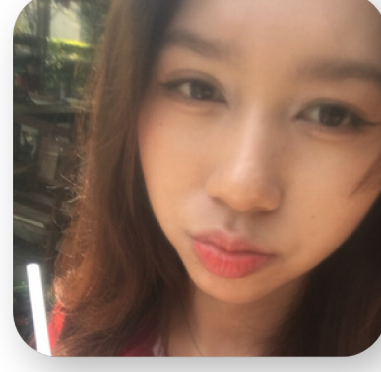
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Overview



Introduction



System Architecture



Feature Engineering



Logic Strategy



Advanced Prompting



The Watcher



Data Splitting Protocol



Backtest result



Test train – factor



System Readiness & Live
integration



Thank you

Overcoming Volatility with Layered Architecture



Problem

- High volatility
- News have high impact on gold price
- Back test value has a discrepancy



Objectives

- Create an automated trading system that integrates Technical + News by making more accurate decisions
- Strong Risk Control
- Ready to trade

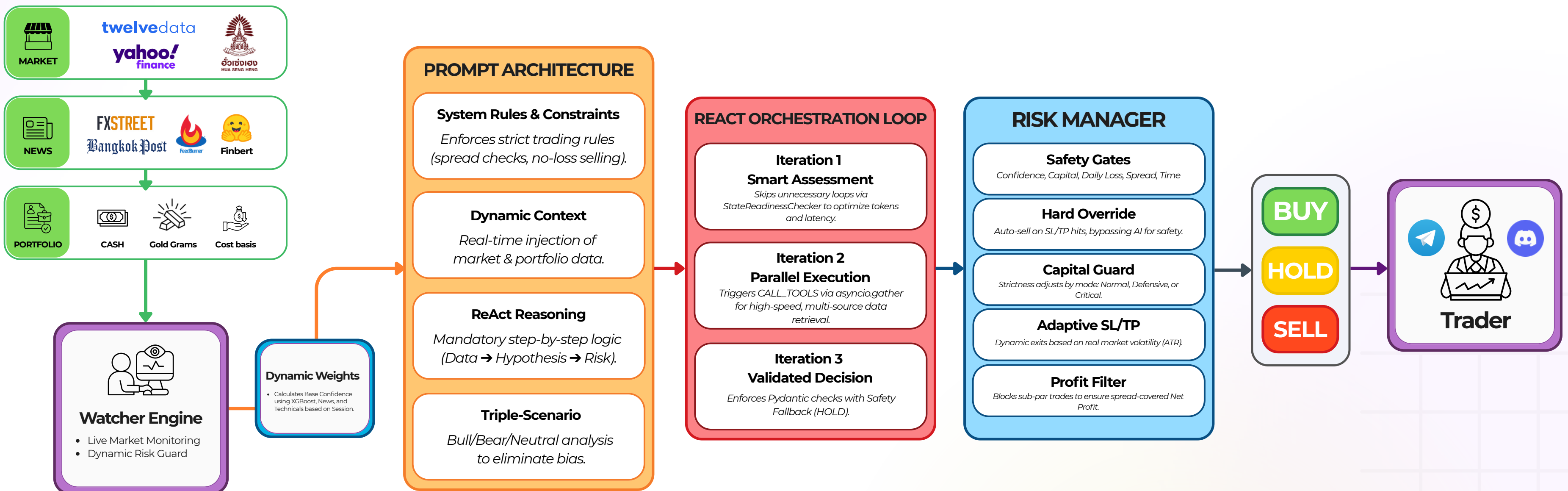


Key Update

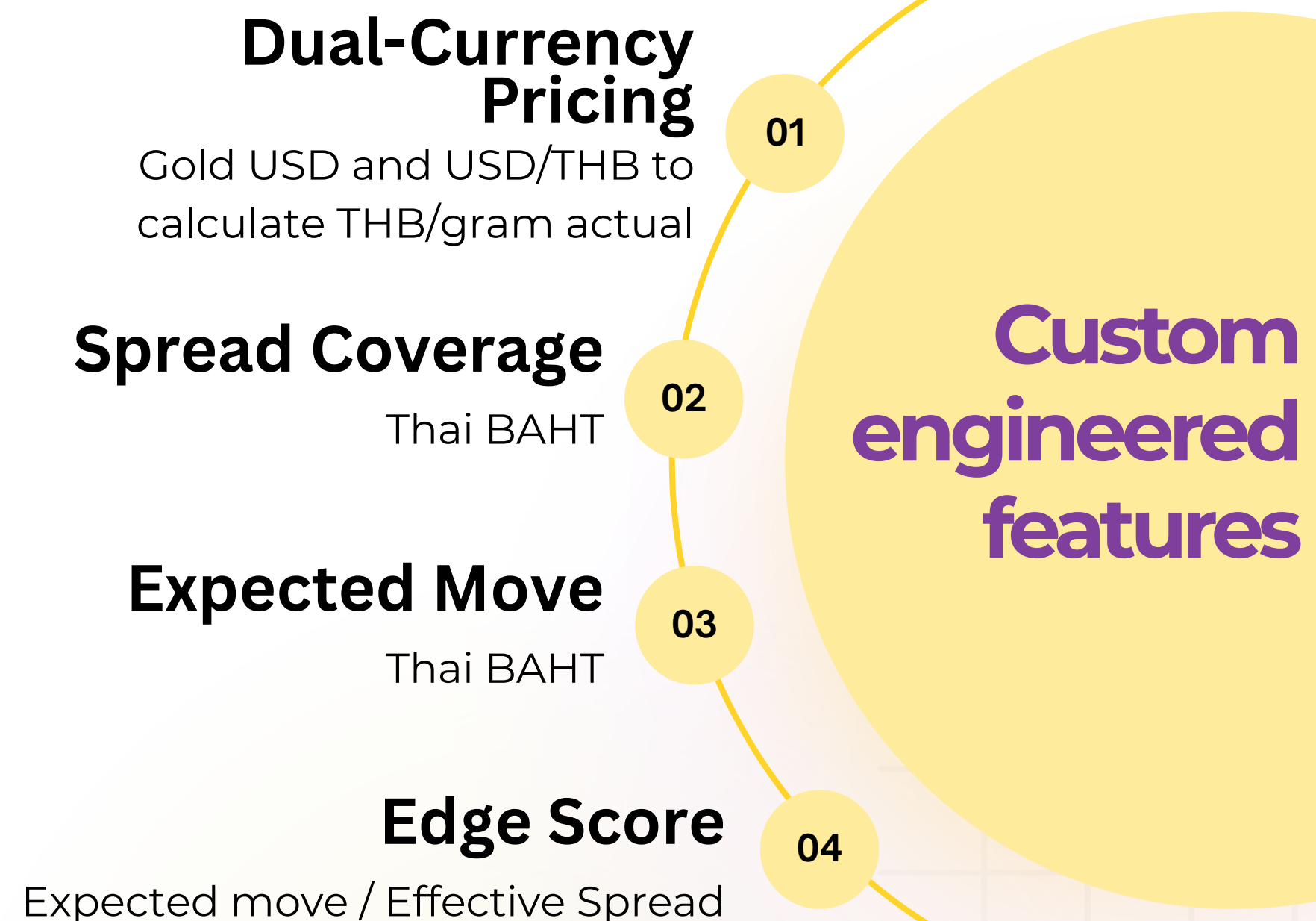
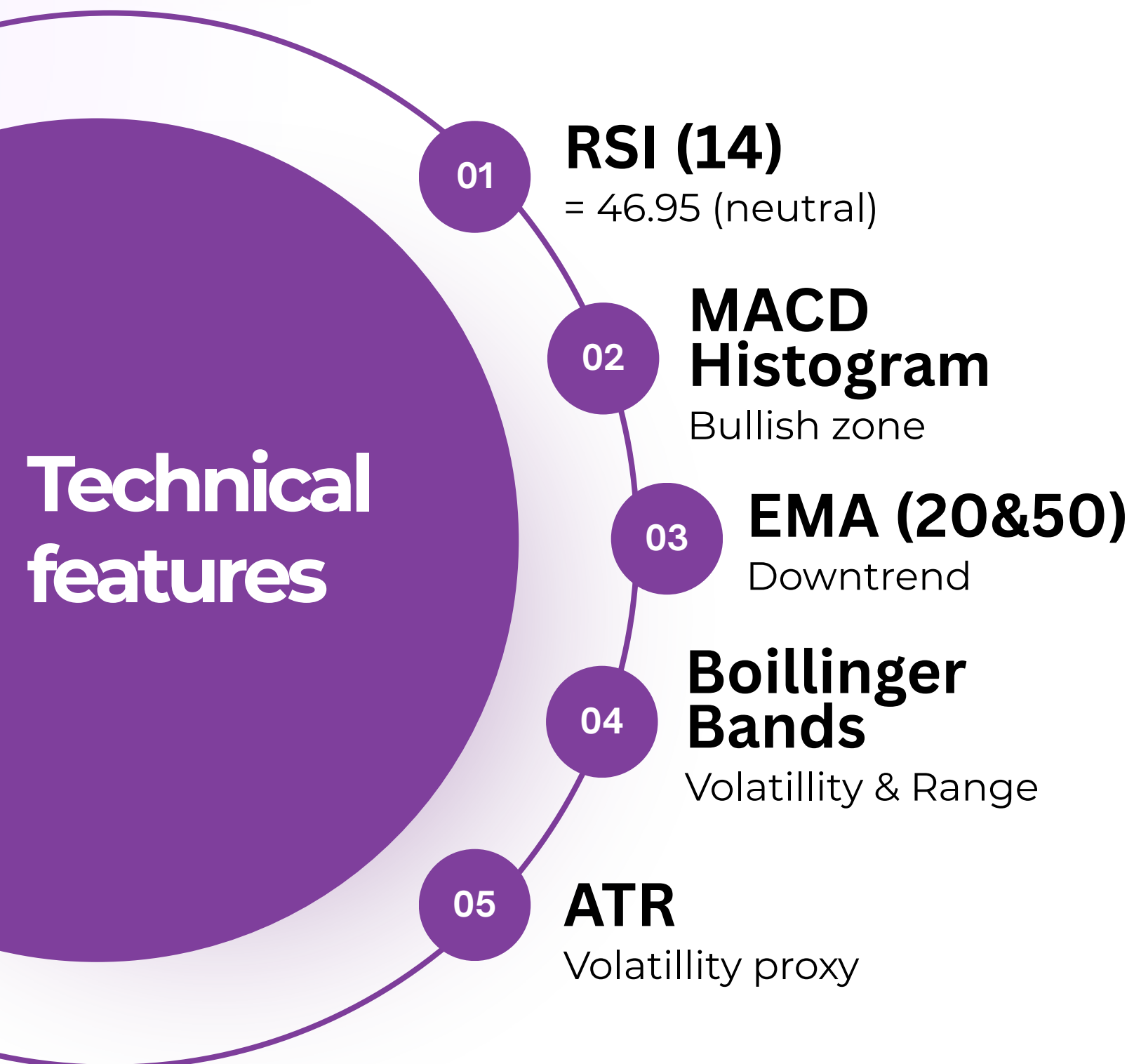
- Improve the structure into Layered Architecture
- Adding Watcher
- Completely connect to the real trading system

System Architecture

End-to-End Layered



Feature Engineering



Logic Strategy

Buy rule

Gate 1: The Gatekeeper (Value)

- **Edge Score ≥ 1.0 ONLY** (Expected move must beat the spread)
- If less than 1.0 $\rightarrow$ HOLD

Gate 2: Technical Setup (Safe Entry)

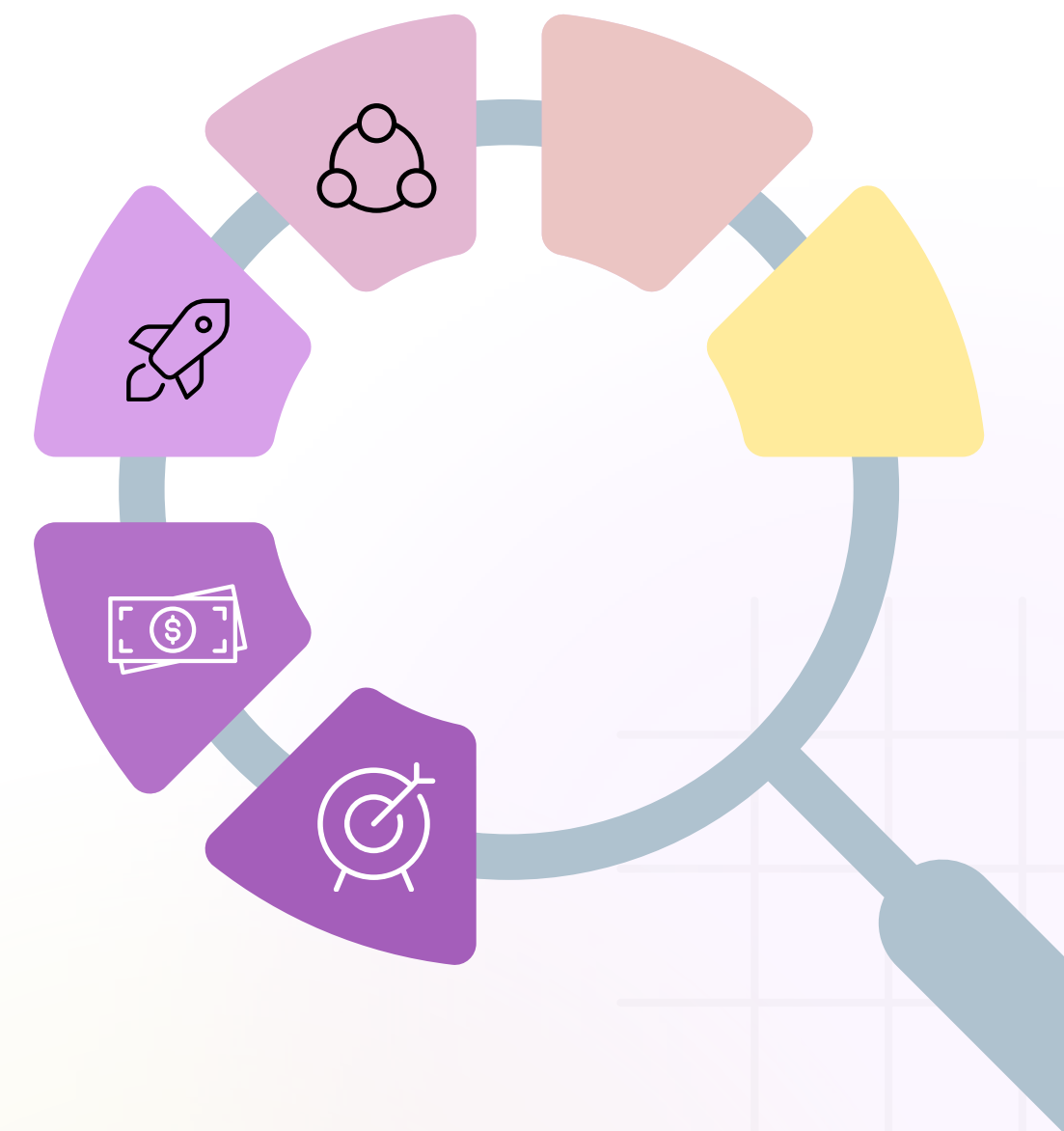
- **Entry Zone** : Price touches the lower Bollinger Band or drops by $\geq 0.2\%$
- **Momentum Hook** : Must be curling up (RSI crossing up from < 30 or MACD Histogram turns positive)
- **Prohibitions** : Absolutely NO buying during flat/sideways markets or RSI > 60 (overbought)

Gate 3: Confidence Calibration

- AI can only adjust the Base Confidence Score by a maximum of ± 0.05 based on Higher Time Frame (HTF) alignment.
- Maximum confidence is capped at 0.85 (unless all factors confirm 100% convergence)

Sell rule

- **Take Profit / Stop Loss**: Hits the target set by the Risk Manager
- **Momentum Fade**: Currently in Profit + Held for > 30 minutes + RSI > 65
- **Early Cut**: Currently in Loss + Held for > 45 minutes + Momentum degrades (MACD dropping or RSI < 52)
- **Hard Exit (Time-based)**: session nearly end— forced liquidation regardless of profit or loss





ADVANCED PROMPTING

Strategy Upgrade

Step 1 (Grounding)



Extract exact numbers
(Price, Spread, PnL,
Cash)

Step 2 (Hypothesis)



Why is the market
moving? Is it Spot USD
or THB currency?

Step 3 (Constraints)



Do I have cash/gold?
Are any exit rules
triggered (e.g., held > 6
candles)?

Step 4 (Edge Gatekeeper)



Does the expected
move cover the
spread?

Step 5 (Weight Calibration)



Anchor to Base
Confidence and adjust
based on skills.



Multi- perspective analysis

json

```
"analysis": {
  "bull_case": "...",
  "bear_case": "...",
  "neutral_case": "..."
}
```

FORCE AI CONSIDER 3 SCENARIOS
PREVENT CONFIRMATION BIAS

Complete Example Output

```
{
  action : FINAL_DECISION
  agent_reasoning : {
    1_data_grounding : Spot=$4626.63, THB=32.47, spread=90.0, PnL=0, held=0, Cash=1500
    2_market_hypothesis : Spot Gold is testing BB lower band support ($4621.87) in a bearish HTF regime.
    3_logical_constraints : No gold held, cannot sell. Must evaluate BUY gates.
    4_edge_gatekeeper : edge_score=2.8474 >= 1.0. Pass.
    5_weight_calibration : Base Conf=0.62. HTF bearish (EMA20 < EMA50) -> -0.05. Final conf=0.57.
    6_momentum_hook_check : RSI=25.96 (not rising). MACD hist=-2.1054 (deepening vs prev -1.2). Gate B fails. FORCE HOLD.
  }
  analysis : {
    bull_case : Price is near BB lower band; potential for mean reversion if RSI hooks up.
    bear_case : HTF trend is bearish; MACD histogram is deepening, indicating continued downward momentum.
    neutral_case : Market is currently in a downtrend without a confirmed momentum reversal.
  }
  execution_check : {
    is_spread_covered : ☒ true
    is_profitable_to_sell : null
  }
  signal : HOLD
  confidence : 0.57
  position_size_thb : null
  rationale : Momentum indicators (RSI/MACD) are failing to hook; MACD histogram is deepening. HTF bearish alignment forces HOLD.
}
```

Tri-Scenario Analysis, Forces the AI to evaluate Bull, Bear, and Neutral cases. Eliminates Confirmation Bias.



THE WATCHER

Fetch market_state

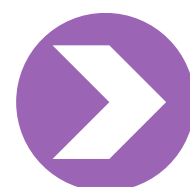
Retrieve market information

data_orchestrator

Read the gold price (Baht / gram)

_extract_price() → THB/GRAM

(None if the information is not enough)



Scenario 1: Holding Gold

DEFENSE Fake Swing Filter

Blocks AI overwhelming for "noise" (long wicks, low momentum, neutral RSI). Prevents unnecessary API costs.

DETECTION Real Reversal Scoring

6-point system (Body ratio, Volume surge, RSI extremes, MAD, ROC flip, Structure break). Requires $\geq 4/6$ to wake AI.

Conditional score

body ratio ≥ 0.5 (close แท่งจริง) → (+1)

volume surge $> 1.3 \times$ average 14 bar → (+1)

RSI extreme (< 30 or > 70) → (+1)

MACD now $> 1.5 \times$ MACD avg (high volatility) → (+1)

ROC flip + → (+1)

structure break (close exceed swing high/low 13 bar before) → (+1)



Scenario 2: Entry & Exit

Take Profit Logic

Triggered when RSI $>$ Overbought threshold or strong overbought signals are detected.

Buy Signal Logic

Triggered when RSI $<$ Oversold threshold or strong oversold signals are detected.

RSI > 70 → **Overbought** → trigger AI

RSI < 30 → **Oversell** → trigger AI

"All signals only *"flag"* the engine; AI makes the *final decision*."



Data Splitting Protocol

Protocol

Split	Date Range	Purpose
Train	1 year	Model development
Test	1 months	Final unseen evaluation

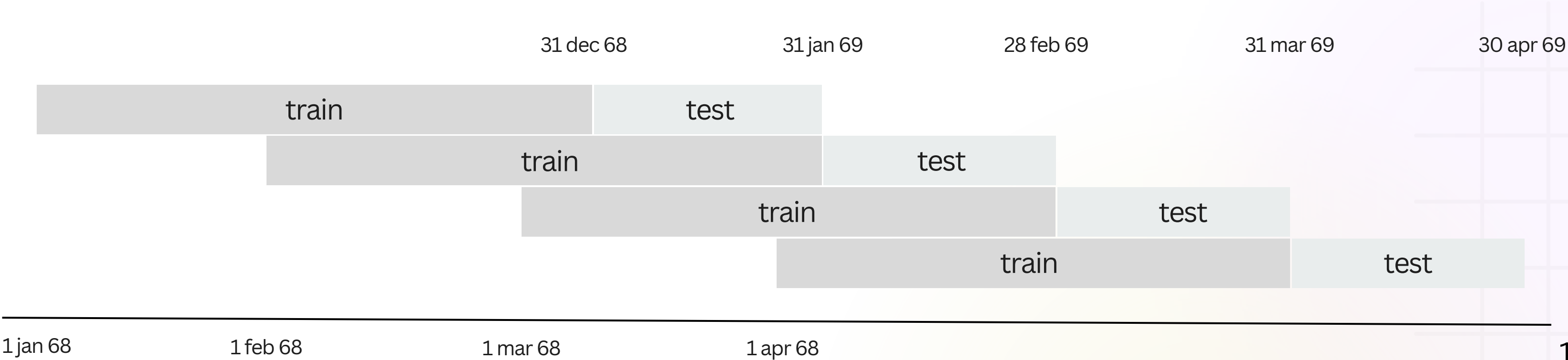
Leakage Prevention

- 

Indicators shifted by 1 bar using shift(1)
- 

No future candle or news used as features
- 

Strict timestamp cutoff across splits





Backtest result

TRADING PERFORMANCE METRICS

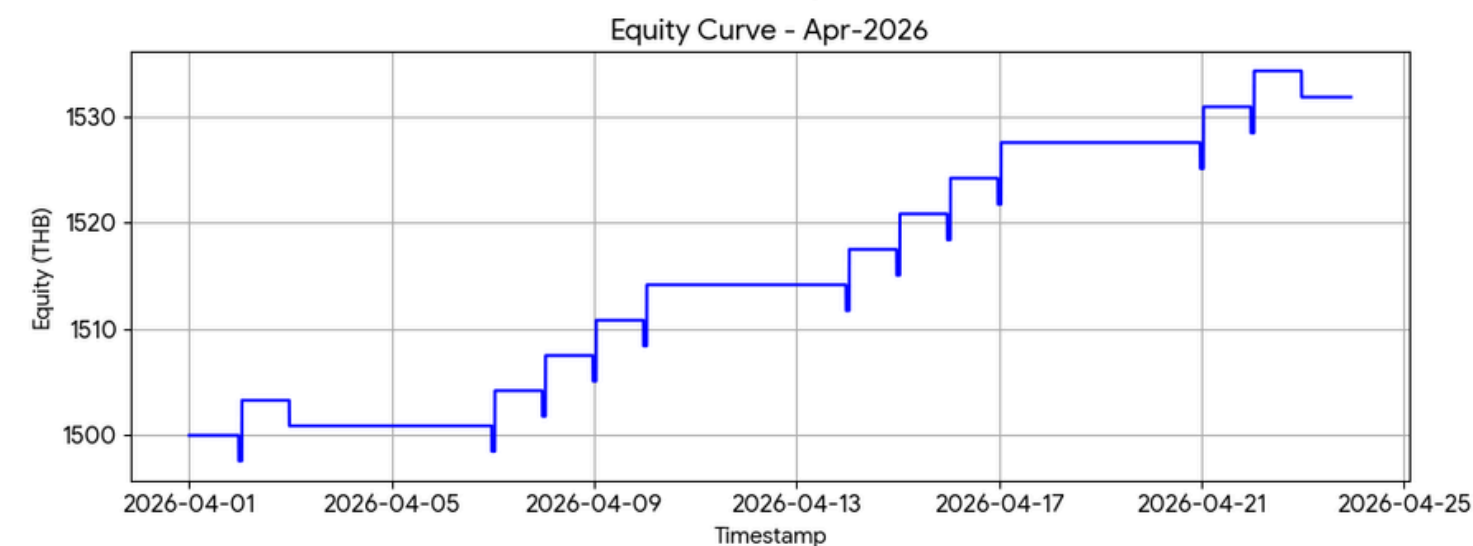
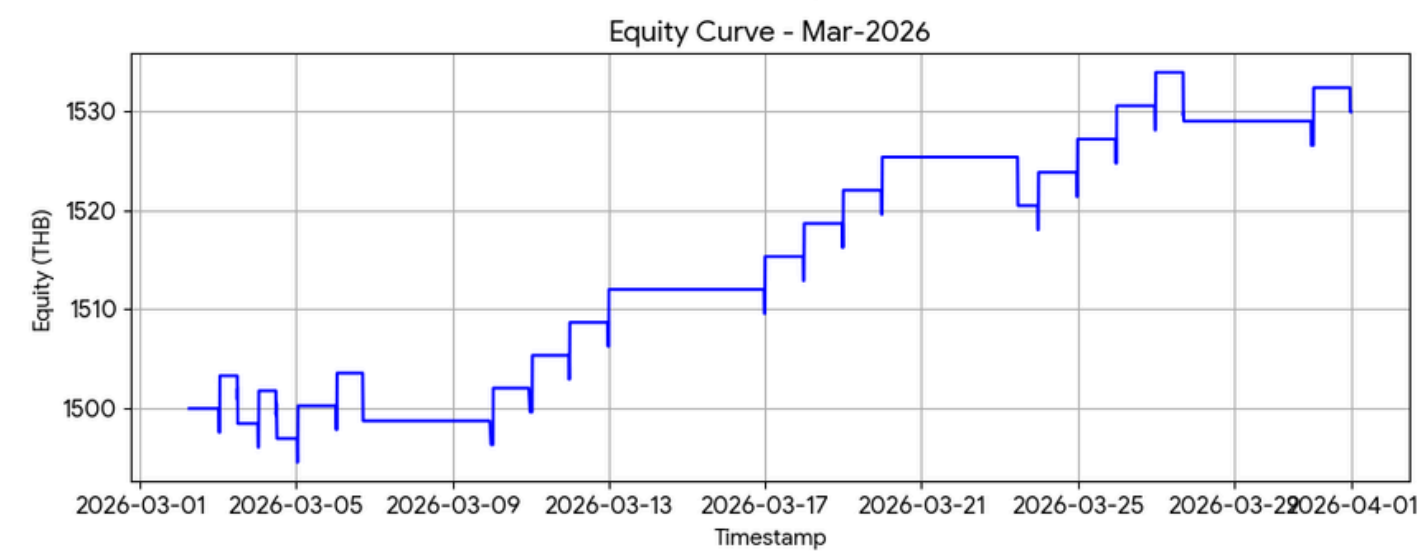
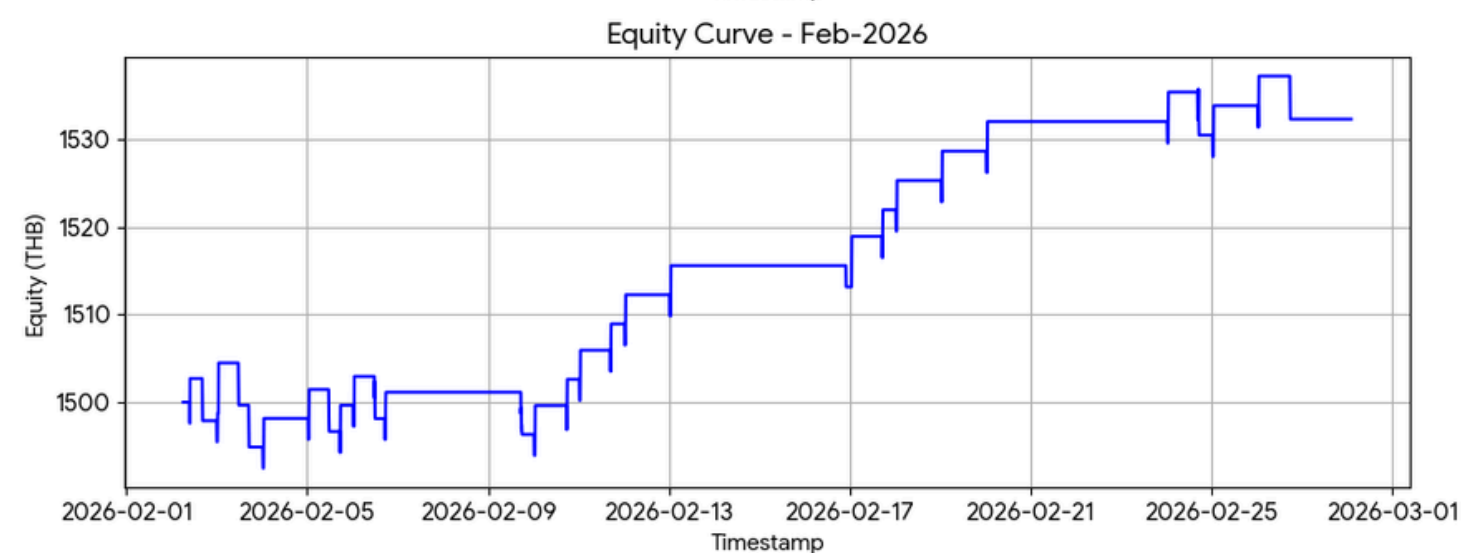
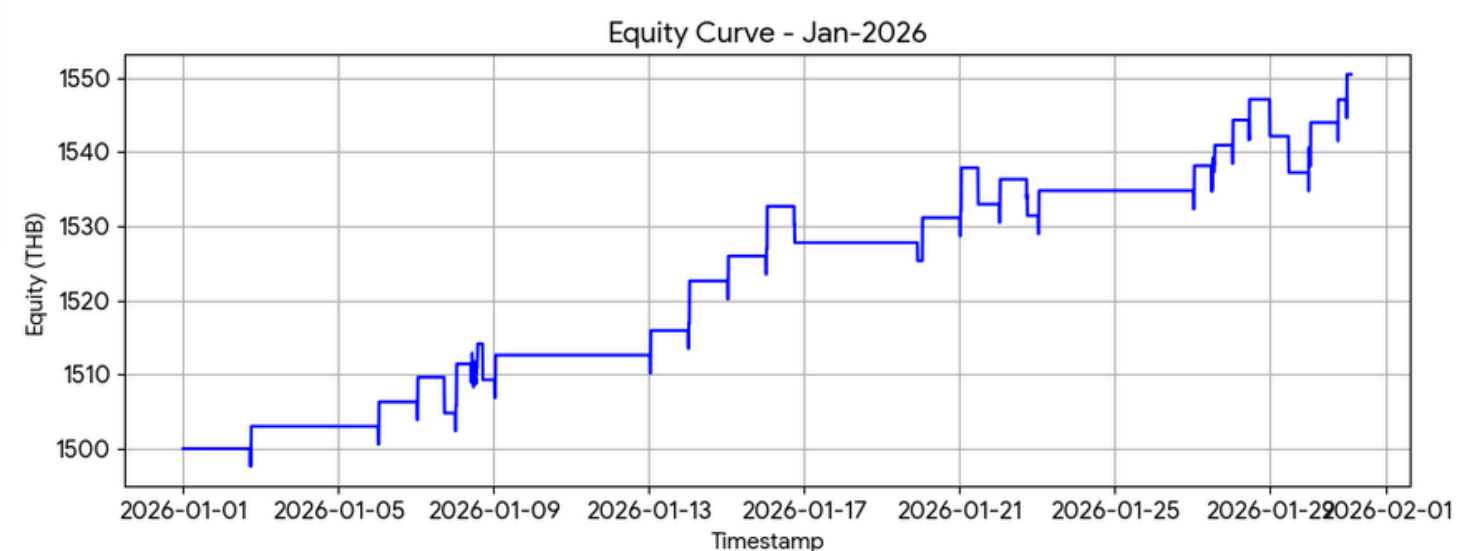
System Backtest & Live Execution Results

	Jan-2026	Feb-2026	Mar-2026	Apr-2026	Combine(Pooled)
Total Closed Trade	33	30	23	13	99
Win/Loss	26/7	22/8	17/6	11/2	76/23
Win Rate (%)	78.79%	73.77%	73.79%	84.62%	76.77%
Total Profit (THB)	50.49	32.31	29.90	31.80	N/A independent ports
Average Win	3.26	3.23	3.33	3.33	3.28
Avarage Loss	4.90	4.83	4.44	2.43	4.54
Expectancy per Trade	1.53	1.08	1.30	2.45	1.46
Best Annualize Trade(%)	2.67	3.11	2.70	3.50	Average annualized CAGR (4 mo) = 33.03%
Worst Annualize Trade(%)	-3.88	-4.52	-3.93	-2.54	
Med Annualize Trade(%)	2.67	3.11	2.70	3.50	
Top 10% Annualize Trade	2.67	3.11	2.70	3.50	
Bottom 10% Annualized Trade	-3.88	-4.52	-3.93	-1.33	
XIRR (%)	49.47	35.17	27.43	39.63	N/A independents ports
Avg Capital/Year (THB/Year)	1500.00 (Independent portfolio. Started at 1500 THB)				
Sharpe Ratio (Month)	0.4403	0.2838	0.3525	1.0817	

*annualized return สูตร CAGR ปกติ ทำมาเพื่อให้ pnl = +3.09฿, r_trade = +0.2020%, days_held = 0.00347 วัน, exponent (ตัวยกกำลัง) = 365/0.00347 = 105,187
 ≈ 1.64 × 10^m %

EQUITY CURVES — Each month independent

System Backtest & Live Execution Results



Objective: To assess performance consistency across various market conditions by isolating monthly results and removing the impact of accumulated profits

TEST TRAIN - FACTOR

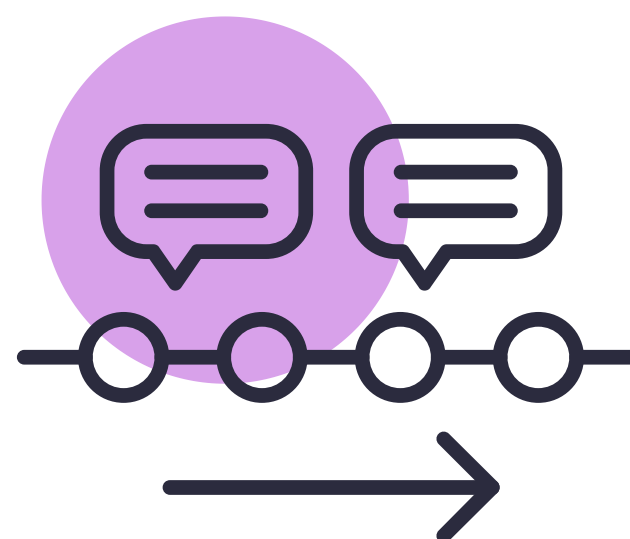
1. Seasonality factors

January - February (The Festive Gap)

- Physical Demand is very high in Asia
 - Chinese New Year
 - Wedding in India
- The price usually has a solid base.
- Often gets a better Win Rate than usual.

April (Songkran Effect - Local Context):

- During Songkran, gold is often sold as a gift.
- Closing positions to hold cash during the long weekend.
- Local market volume is volatile, but in the global market, it will be more based on macro factors.



2. Macroeconomic factors

March (The Pivot Month)

- the most volatile month in the first quarter
- There is a quarter closing (Quarter End)
- There is a central bank (Fed) meeting determines the direction of interest rates in the middle of the year.

Inflation Outlook

- In 2026, the economy may begins to recover
- Gold may be sold for profit to turn money into risky assets
- Variation occurs at the Follow Trend trading system may affected

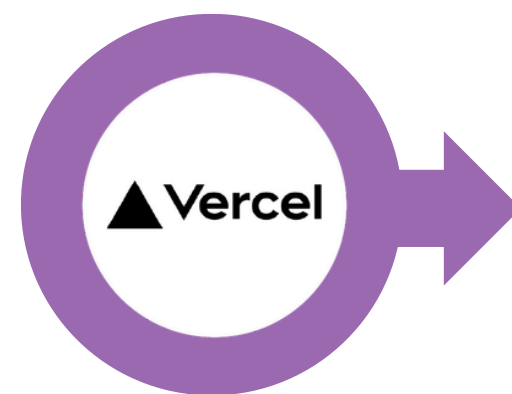
System Readiness & Live integration



01

App Gold now

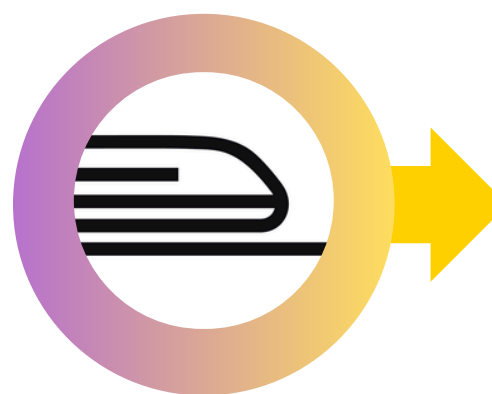
Primary access point via Hua Seng Heng's 'Gold Now' mobile application for users.



02

Vercel Frontend

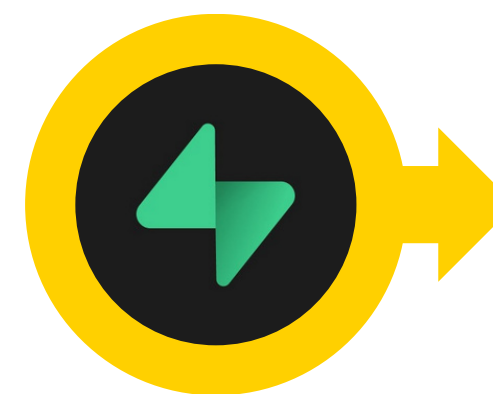
A website system that supports transaction management, data display, and effective order submission



03

Railway

The backend system for trading is responsible for processing all logic, conducting trading and managing control



04

Supabase

The main database is centralized and has an archive for safe storage of all trading history and important information



05

Back Test

The process of simulating trading strategies with historical data to prove the stability of profits and risk management efficiency before putting AI into practice



Thank you